

SKANDESH GOUTHAM RANGANATHAN J

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[LinkedIn](#) | [LeetCode](#) | [GitHub](#)

EDUCATION

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science (CGPA: 8.32)

Aug 2023 – May 2027

Chennai, India

TECHNICAL SKILLS

- **Languages:** Python, C, Java, HTML/CSS, SQL
- **Frameworks:** FastAPI, Flask, Scikit learn, TensorFlow, XGBoost, Streamlit
- **Tools:** Git, AWS, MySQL, Docker, Kali Linux, Postman, Cisco Packet Tracer
- **Concepts:** Data Structures, Object oriented Design, Database Management RSA & SHA-256, APIs, Computer Networks, Cloud Computing

KEY PROJECTS

Plumb: Context-Aware Ball Tracking System *Python, XGBoost, Streamlit, Docker* [GitHub](#)

- Architected a 15-module context-aware intelligence pipeline fusing ball-tracking data (HawkeyeStats) with bowler biomechanics, pitch, venue, and live weather (Open-Meteo API) signals, using XGBoost and a custom physics simulation to model expected delivery behavior beyond raw trajectory reconstruction.
- Flagged anomalies by comparing predicted versus observed outcomes to power wide-ball detection and no-ball classification, and applied SHAP-based explainability to validate feature contributions via an interactive Streamlit dashboard containerized with Docker Compose and Nginx.

Cloud Load Forecaster (HDRL-LF) *Python, TensorFlow, LSTM* [GitHub](#)

- Engineered a hybrid VM placement framework processing 10k+ data points, achieving 92% forecasting accuracy to optimize cloud resource allocation.
- Enhanced system energy efficiency by 15% through the implementation of Dueling Deep Q-Network (DQN) agents for dynamic load balancing.

Vault: AI Financial Management *Flask, Python* [GitHub](#)

- Developed a full-stack financial application featuring automated transaction tracking and a regression-based engine to forecast spending, improving prediction accuracy by **[91]%**.
- Integrated a conversational AI assistant (Penny) with gamified alerts, increasing user engagement by **[90]%** and reducing budget overruns by **[45]%** through data-driven alerts.

SecureChain: Digital Evidence Custody *FastAPI, PostgreSQL, SHA-256* [GitHub](#)

- Architected a cybersecurity platform using Dual-Chain Hashing (DCH-IV) to ensure 100% integrity of digital evidence without blockchain overhead.
- Engineered a cross-chain binding mechanism with RBAC and JWT authentication, reducing forensic verification latency by 40% across RESTful nodes.

OfferCred: Certificate Verification *Node.js, Express, PostgreSQL* [GitHub](#)

- Built an RSA-signature platform that authenticated 200+ certificates and successfully blocked 50+ fraud attempts using custom validation logic.
- Implemented Bcrypt and JWT-based security protocols, resulting in a 99% reduction in unauthorized system access incidents during testing phases.

PUBLICATIONS & RESEARCH

NeuroBridge - AI Mental Health Crisis Detection

- Proposed a federated AI system for real-time crisis detection using multi-device behavioral drift analysis.

Autonomous Vehicle Security

- Analyzed vulnerabilities in autonomous systems and proposed software-based defenses against ethical hacking.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **AWS Academy Graduate:** Gained hands-on experience in AWS Cloud architecture and storage via AWS Student Platform labs.
- **Networking Basics (Cisco):** Specialized in TCP/IP protocols and connectivity troubleshooting using Cisco Packet Tracer.
- **Cloud Computing (NPTEL):** Trained in Virtualization, Resource Management, and SLA-based scheduling in distributed systems.
- **Ethical Hacking (NPTEL):** Conducted network security audits and penetration testing within **Kali Linux** environments.
- **Machine Learning (Coursera):** Applied Scikit-Learn for Supervised/Unsupervised Learning and Model Tuning.
- **Industry Simulations (Forge):**
 - Completed AWS Solutions Architecture by designing scalable and resilient cloud environments for key client services.
 - Completed Deloitte Data Analytics by using Python to analyze and visualize large datasets for business intelligence.